

Prime-Control Lattices

A Prime-Coordinate Toy Universe: Multiplicative Coding, Beat Channels, and Quantum Control

Won Sik Choi

Email: janefather@gmail.com

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Abstract

We introduce a self-contained mathematical toy model in which a finite set of prime numbers indexes coordinate directions of a multiplicative state space. The underlying group is a free abelian lattice, but the group alone does not canonically select the prime basis: unimodular changes of basis generate equivalent multiplicative coordinate systems. We therefore distinguish the nonnegative-exponent multiplicative monoid. Its atoms are exactly the selected primes, and its automorphism group reduces to permutations of those atoms, giving a precise prime-selection principle once an oriented factorization monoid is part of the structure. Bilateral prime shifts are symmetrized to define inversion-preserving control operators. Their product obeys an exact product-ratio mixing identity, while an m -prime product generates 2^{m-1} inversion-identified channels: one forward compositional code channel and $2^{m-1}-1$ mixed-sign beat channels. A logarithmic position observable converts multiplicative coordinates into additive scale, and a quadratic logarithmic drift gives an exact bare splitting $\Omega \log p \log q$ between product and ratio states. For finite hard-wall truncations we prove full unitary controllability inside each inversion-parity sector under independently addressable prime-axis controls. Inversion symmetrization also erases orientation, so a physical realization would need an additional preparation or positive-monoid structure to identify which direction is primitive-forward. The construction is a mathematical control architecture, not a claim that physical nature is prime-coded.

Keywords: prime-exponent lattice; positive-exponent monoid; basis gauge; prime selection; multiplicative state space; quantum control; quantum walk; Lie algebra; controllability; beat channel; logarithmic dynamics

Scope and independence. This preprint is self-contained. It does not rely on the Riemann hypothesis, zeta-zero dynamics, or any prior arithmetic-filter construction. Its mathematical lineage is instead through unique factorization and Bohr-type prime multi-index coordinates, lattice quantum dynamics, and Lie-algebraic quantum controllability.

1. Introduction

Prime factorization converts multiplication into integer-coordinate addition. For a fixed finite set of distinct primes, the multiplicative subgroup that they generate is therefore a copy of an integer lattice. This elementary observation is closely related to the prime multi-index viewpoint underlying the Bohr lift for Dirichlet series, where an integer is mapped to the vector of its prime exponents [1,2]. In a different literature, continuous-time quantum walks use graph or lattice adjacency operators as Hamiltonian components [3,4], while finite-dimensional quantum control characterizes reachable unitary dynamics through the Lie algebra generated by drift and control Hamiltonians [5–7].

The purpose of this paper is to combine these ingredients into one minimal control model. The model treats prime exponents as coordinates, assigns a symmetric shift control to each prime axis, and uses a quadratic logarithmic drift to make the controls dynamically noncommuting. The resulting algebra has two notable features. First, multiplying symmetric prime controls produces an exact pair of product and ratio channels. Second, after finite truncation, the system is fully controllable within each inversion-parity sector.

No claim is made that prime coordinates are physically fundamental. The abstract multiplicative group alone does not select a unique basis: after identifying the group with $\mathbb{Z}^m$, any unimodular basis change in $GL(m, \mathbb{Z})$ gives another valid multiplicative coordinate system. Prime selection becomes canonical only after one also distinguishes the nonnegative-exponent integer monoid, whose atoms are the prime directions. The product-ratio identity itself still extends to arbitrary multiplicatively independent generators. The model should therefore be read as a mathematically explicit toy universe in which the distinction between group-level coordinate freedom and monoid-level primitive atoms is made explicit.

2. Multiplicative State Space, Basis Gauge, and Prime Selection

Let $P = \{p_1, \dots, p_m\}$ be a finite set of distinct primes. Define the multiplicative subgroup

$$\Gamma_P := \{ \prod_{j=1}^m p_j^{k_j} : k = (k_1, \dots, k_m) \in \mathbb{Z}^m \} \subset \mathbb{Q}_{>0}^{\times}.$$

Proposition 1 (prime-coordinate isomorphism). The map $\chi: \mathbb{Z}^m \rightarrow \Gamma_P$ given by $\chi(k) = \prod_{j=1}^m p_j^{k_j}$ is a group isomorphism from $(\mathbb{Z}^m, +)$ to $(\Gamma_P, \times)$.

Proof. The map is a homomorphism because exponents add under multiplication. Surjectivity holds by definition. If $\chi(k) = 1$, then $\prod_{j=1}^m p_j^{k_j} = 1$. Clearing negative exponents gives equality between two positive integers with disjoint prime factorizations; unique factorization forces $k_j = 0$ for every j . Hence χ is injective. ■

For the full set of primes this becomes the standard free-abelian decomposition $\mathbb{Q}_{>0}^{\times} \cong \prod_p \mathbb{Z}$. This group representation is useful, but by itself it does not make the prime basis unique.

2.1. Basis-gauge obstruction

Let $A \in GL(m, \mathbb{Z})$. Define new multiplicative generators g_j by $g_j = \prod_{i=1}^m p_i^{A_{ij}}$. Because A is unimodular, the exponent transformation is invertible over $\mathbb{Z}$, so $\{g_1, \dots, g_m\}$ is another ordered group basis of Γ_P .

Proposition 2 (basis-gauge obstruction). The abstract group Γ_P does not canonically identify the prime axes. Every $A \in GL(m, \mathbb{Z})$ produces an alternative multiplicative group basis, and every ordered group basis arises from such an integer-unimodular change of coordinates.

Proof. Under Proposition 1, Γ_P is identified with $\mathbb{Z}^m$. Ordered bases of the free abelian group $\mathbb{Z}^m$ are exactly the images of the standard basis under $GL(m, \mathbb{Z})$. Transporting such a basis through χ gives the stated multiplicative generators. ■

For example, in $\langle 2, 3 \rangle$ the pair $\{2, 6\}$ is a valid group basis because $6 = 2 \cdot 3$ and $3 = 6/2$.

2.2. The nonnegative-exponent monoid

To select a forward factorization structure, distinguish the submonoid

$$M_P := \{ \prod_{j=1}^m p_j^{n_j} : n_j \in \mathbb{N}_0 \} \cong \mathbb{N}_0^m.$$

An atom of a commutative monoid is a non-identity element that cannot be written as a product of two non-identity elements.

Theorem 1 (prime selection from the positive-exponent monoid). The atoms of M_P are exactly $p_1, \dots, p_m$. Every monoid automorphism of M_P permutes these atoms; equivalently $\text{Aut}(M_P) \cong S_m$. Hence, once M_P is distinguished inside Γ_P , the prime axes are unique up to permutation.

Proof. Under $M_P \cong \mathbb{N}_0^m$, the nonzero indecomposable elements under addition are exactly the standard basis vectors $e_1, \dots, e_m$: any vector with coordinate sum at least two decomposes into two nonzero vectors. Thus the atoms are precisely e_j , corresponding to p_j . Any monoid automorphism preserves atomicity and must therefore permute the e_j ; conversely every permutation is an automorphism. ■

This separates two notions that are otherwise easy to conflate: Γ_P supplies multiplicative coordinate freedom, whereas M_P supplies a distinguished factorization orientation and selects primes as primitive atoms.

3. Symmetric Prime Shift Operators

Work first on the infinite Hilbert space $\mathcal{H}_P = \ell^2(\mathbb{Z}^m)$, with orthonormal basis $\{|k\rangle : k \in \mathbb{Z}^m\}$. For the j -th prime define the bilateral unitary shift

$$S_j |k\rangle = |k + e_j\rangle, \quad S_j^\dagger |k\rangle = |k - e_j\rangle.$$

The inversion-symmetric prime control is

$$Q_j := \frac{1}{2}(S_j + S_j^\dagger).$$

More generally, for $x = \prod_{j=1}^m p_j^{r_j} \in \Gamma_P$ define $S_x := \prod_{j=1}^m S_j^{r_j}$ and

$$Q_x := \frac{1}{2}(S_x + S_x^\dagger).$$

Orientation loss under inversion symmetrization. Since $Q_x = Q_{1/x}$, the symmetric control remembers the unoriented multiplicative axis but not which direction belongs to M_P . Thus the symmetric dynamics alone cannot reconstruct the

forward monoid: a preparation convention, positive-exponent cone, or equivalent orientation datum must be supplied separately if one wants prime atoms to be operationally identifiable.

Because the shifts commute, the Q_x form a simple commutative mixing algebra even though they will not commute with the drift introduced below.

4. Product-Ratio Mixing and the Prime Hypercube Law

Theorem 2 (product-ratio mixing). For all $x, y \in \Gamma_P$, $Q_x Q_y = \frac{1}{2}(Q_{\{xy\}} + Q_{\{x/y\}})$.

Proof. Expand $Q_x Q_y = \frac{1}{4}(S_x + S_x^\dagger)(S_y + S_y^\dagger)$. Since $S_x S_y = S_{\{xy\}}$, $S_x S_y^\dagger = S_{\{x/y\}}$, and $Q_z = Q_{\{1/z\}}$, the four terms pair into $Q_{\{xy\}}$ and $Q_{\{x/y\}}$. ■

For distinct primes p and q ,

$$Q_p Q_q = \frac{1}{2}(Q_{\{pq\}} + Q_{\{p/q\}}).$$

The first term is a product channel and the second is a ratio, or beat, channel. For the pair 2 and 3 this gives

$$Q_2 Q_3 = \frac{1}{2}(Q_6 + Q_{\{3/2\}}).$$

Theorem 3 (prime hypercube law). For distinct primes $p_1, \dots, p_m$, the product $\prod_{j=1}^m Q_{\{p_j\}}$ contains exactly $2^{(m-1)}$ inversion-identified channels:

$$\prod_{j=1}^m Q_{\{p_j\}} = 2^{(1-m)} \sum_{\{\epsilon \in \{\pm 1\}^m / (\epsilon \sim -\epsilon)\}} Q_{\{\prod_{j=1}^m p_j^{\epsilon_j}\}}.$$

Proof. Expanding $\prod_j (S_{\{p_j\}} + S_{\{p_j\}}^\dagger)$ gives 2^m signed products. Global sign reversal sends x to $1/x$, and $Q_x = Q_{\{1/x\}}$; hence the 2^m terms form $2^{(m-1)}$ pairs. Distinctness follows from unique factorization. ■

For three primes 2, 3, 5 the four inversion-identified channels may be represented by 30, 6/5, 10/3, and 15/2. The exponential channel count is algebraic; it does not depend on a fitted dynamical parameter.

Corollary 1 (hypercube code-beat split). Relative to the distinguished monoid M_P , among the $2^{(m-1)}$ inversion-identified channels generated by distinct primes $p_1, \dots, p_m$, exactly one class has a representative in M_P : the product channel $p_1 p_2 \cdots p_m$. Every other class has mixed-sign exponents and lies neither in M_P nor in $M_P^{\{-1\}}$. Hence the hypercube decomposes into one forward compositional code channel and $2^{(m-1)} - 1$ mixed-sign beat channels.

Proof. Each hypercube term has exponent vector $\epsilon \in \{\pm 1\}^m$, modulo global sign $\epsilon \sim -\epsilon$. A class has a nonnegative representative exactly when all signs agree. That single class is represented by $(+1, \dots, +1)$, which gives $p_1 \cdots p_m$. Any other class contains both $+1$ and -1 entries, and global sign reversal preserves the mixed-sign property. ■

For 2, 3, 5, the channel 30 is the unique forward compositional codeword, while 6/5, 10/3, and 15/2 are mixed-sign beat channels.

5. Logarithmic Coordinate and Quadratic Drift

Set

$$a_j := \frac{1}{2} \log p_j, \quad U|k\rangle := u(k)|k\rangle, \quad u(k) := \sum_{j=1}^m a_j k_j.$$

Under the multiplicative identification $x = \chi(k)$, the observable U is simply $U = (1/2) \log x$ on basis states. The shift relation is

$$[U, S_j] = a_j S_j.$$

We use the quadratic logarithmic drift

$$D := \Omega U^2, \quad \Omega \neq 0.$$

Since $[U^2, Q_j] \neq 0$, prime-coordinate controls are dynamically active rather than merely relabeling states. A convenient quantum-walk-style Hamiltonian is

$$H_0 := \Omega U^2 + \kappa \sum_{j=1}^m (2I - S_j - S_j^\dagger),$$

with controlled dynamics

$$H(t) := H_0 + \sum_{j=1}^m g_j(t) Q_j.$$

Because $S_j + S_j^\dagger = 2Q_j$, the hopping term in H_0 is a linear combination of the same independently available controls plus the identity. Consequently, the projective controllability analysis is equivalent to that of the generator set $\{iU^2, iQ_1, \dots, iQ_m\}$.

6. A Bare Prime-Pair Clock

For two primes $p < q$, consider the product basis state pq and the ratio basis state q/p . Their logarithmic coordinates are

$$u_+ = \frac{1}{2}\log(pq), \quad u_- = \frac{1}{2}\log(q/p).$$

Proposition 3 (quadratic-drift pair splitting). Under the bare drift $D = \Omega U^2$, the energy difference between these two basis states is

$$E_{\{pq\}} - E_{\{q/p\}} = \Omega \log p \cdot \log q.$$

Proof. Use $E = \Omega u^2$ and the identity $(A+B)^2 - (B-A)^2 = 4AB$ with $A = (1/2)\log p$ and $B = (1/2)\log q$. ■

For $p=2$ and $q=3$, the bare splitting is approximately 0.7615000104Ω . This formula is exact for the diagonal quadratic drift. When hopping is included, the product and ratio basis states are generally not eigenstates of the full Hamiltonian, so the expression should not be interpreted as an exact dressed spectral gap.

7. Finite Hard-Wall Prime Lattices

For control theory we truncate to the inversion-symmetric box $\Lambda_K = \{-K, \dots, K\}^m$. Let $\mathcal{H}_K$ be the span of $\{|k\rangle; k \in \Lambda_K\}$. Define hard-wall shifts T_{j^+} by $T_{j^+}|k\rangle = |k+e_j\rangle$ when $k+e_j \in \Lambda_K$ and zero otherwise, and set $T_{j^-} = (T_{j^+})^\dagger$. Then

$$Q_j(K) := \frac{1}{2}(T_{j^+} + T_{j^-}), \quad U_K|k\rangle := u(k)|k\rangle.$$

The finite controlled Hamiltonian is

$$H_K(t) = \Omega U_K^2 + \kappa \sum_j (2I - T_{j^+} - T_{j^-}) + \sum_j g_j(t) Q_j(K).$$

The exact product-ratio group identity belongs to the infinite group algebra. Hard-wall truncation modifies products at the boundary, but the finite model retains the prime-axis controls, logarithmic drift, and inversion symmetry needed for the controllability result.

8. Inversion Symmetry and Parity Sectors

Define the inversion operator

$$\Pi|k\rangle := |-k\rangle.$$

Proposition 4 (parity conservation). On Λ_K , Π commutes with U_K^2 and with every $Q_j(K)$. Hence $\mathcal{H}_K$ splits into invariant even and odd sectors $\mathcal{H}_+ \oplus \mathcal{H}_-$.

Proof. The equality $u(-k)^2 = u(k)^2$ gives $[\Pi, U_K^2] = 0$. Inversion exchanges T_{j^+} and T_{j^-} , so it fixes their symmetric sum. ■

The only inversion-fixed lattice point is the origin. With $N = (2K+1)^m$, the sector dimensions are therefore

$$d_+ = (N+1)/2, \quad d_- = (N-1)/2.$$

Thus the model cannot mix the two parity sectors using only inversion-symmetric prime controls. The relevant question is whether control is complete inside each permitted sector.

9. Sector Controllability Theorem

We now prove that the answer is yes for every finite prime set and every $K \geq 1$, provided each prime axis is independently addressable.

Lemma 1 (rational independence of prime logarithms). If $p_1, \dots, p_m$ are distinct primes and $\sum_j c_j \log p_j = 0$ with $c_j \in \mathbb{Q}$, then $c_j = 0$ for every j .

Proof. Multiply by a common denominator to obtain integer coefficients n_j . Exponentiation gives $\prod_j p_j^{n_j} = 1$, and unique factorization forces all $n_j = 0$. ■

For a directed j -edge $k \rightarrow k+e_j$ inside Λ_K , the quadratic-drift transition gap is

$$\omega_j(k) = \Omega[(u(k)+a_j)^2 - u(k)^2] = \Omega a_j[2u(k)+a_j].$$

Lemma 2 (gap separation within a controlled prime axis). For fixed j , $|\omega_j(k)| = |\omega_j(\ell)|$ if and only if either $\ell = k$ or $\ell = -k - e_j$, the latter being the inversion-reflected edge.

Proof. If the gaps have the same sign, $u(k)=u(\ell)$, and Lemma 1 gives $k=\ell$. If they have opposite sign, $u(k)+u(\ell)+a_j=0$, equivalently $\sum_r a_r(k_r+\ell_r+\delta_{rj})=0$; Lemma 1 gives $\ell=-k-e_j$. The gap cannot vanish because that would require the odd integer $2k_j+1$ to vanish. ■

Theorem 4 (full controllability in each inversion-parity sector). Let P be any finite set of distinct primes, $K \geq 1$, and $\Omega \neq 0$. Consider the bilinear finite-dimensional system generated by the drift U_K^2 and the independently controlled symmetric prime-axis operators $Q_1^\wedge(K), \dots, Q_m^\wedge(K)$. Restricted to either nonzero inversion-parity sector, the dynamical Lie algebra contains $\mathfrak{su}(d_\pm)$; with the nonzero trace of U_K^2 it is $\mathfrak{u}(d_\pm)$.

Equivalently, the system is fully unitary controllable within each parity sector.

Proof. Restrict first to one parity sector. The drift is diagonal in the parity-orbit basis because k and $-k$ have equal energy. For a fixed prime direction j , $Q_j^\wedge(K)$ is a sum of nonzero symmetric couplings over inversion-orbits of j -edges. By Lemma 2, the squared transition gaps of these edge orbits are distinct. Therefore repeated application of $\text{ad}_{\{iU_K^2\}^2}$ to $iQ_j^\wedge(K)$, followed by finite Lagrange interpolation in the distinct squared gaps, isolates the symmetric generator on any chosen j -edge orbit. A further commutator with iU_K^2 produces the corresponding antisymmetric edge generator because every gap is nonzero. Doing this for all j supplies both standard skew-Hermitian edge generators on the parity quotient graph. The even quotient graph is connected because it is the quotient of the connected box lattice. The odd quotient graph is also connected: for $m \geq 2$ the punctured box $\Lambda_K \setminus \{0\}$ is connected, while for $m=1$ its two components are exchanged and identified by inversion. On a connected graph, symmetric and antisymmetric generators on the edges generate $\mathfrak{su}(d)$ by commutators along paths and by edge commutators that produce diagonal differences. Finally U_K^2 has nonzero trace on each nonzero sector, so adjoining its scalar component gives $\mathfrak{u}(d)$. ■

The theorem is stronger than a numerical rank observation: it applies to arbitrary finite collections of distinct primes and arbitrary hard-wall size K . It also clarifies what is and is not specifically prime-dependent. The controllability proof uses the rational independence of $\log p_j$, a property shared by any multiplicatively independent rational generators. Accordingly, the abstract group-level dynamics does not by itself select the prime basis. Prime selection enters through the distinguished positive-exponent monoid M_P , whose atoms are exactly the prime directions and are unique up to permutation.

10. Minimal 2-3 Toy Universe

Take $P=\{2,3\}$. A basis state is $|k_2, k_3\rangle$ and

$$U|k_2, k_3\rangle = \frac{1}{2}(k_2 \log 2 + k_3 \log 3)|k_2, k_3\rangle.$$

The identity state is $|0,0\rangle$. The product state 6 corresponds to $(1,1)$, while the beat state $3/2$ corresponds to $(-1,1)$. Their inversion partners represent $1/6$ and $2/3$. The symmetric controls obey

$$Q_2 Q_3 |0,0\rangle = \frac{1}{4}(|1,1\rangle + |1,-1\rangle + |-1,1\rangle + |-1,-1\rangle),$$

which groups into equal product and ratio parity channels. Relative to $M_{\{2,3\}}$, 6 is the forward product codeword while $3/2$ is a mixed-sign beat coordinate. For $K=1,2,3$, the parity-sector dimensions and the corresponding projective Lie-algebra dimensions predicted by Theorem 4 are:

K	Total N	d_+	d_-	$\dim \mathfrak{su}(d_+)$	$\dim \mathfrak{su}(d_-)$
1	9	5	4	24	15
2	25	13	12	168	143
3	49	25	24	624	575

Including physically irrelevant global phase, the full sector Lie algebras are $\mathfrak{u}(d_\pm)$ with dimensions $d_\pm \pm 2$. Hence any normalized pure state in a chosen parity sector can, in principle, be connected to any other by an admissible control sequence.

11. Interpretation as Multiplicative Coding

The model admits a compact coding interpretation with two layers. At the group level, multiplicative coordinates have a $GL(m, \mathbb{Z})$ basis gauge. At the oriented monoid level, primes are the atoms of M_P and are unique up to permutation. A composite integer is a forward composition of these atoms; inversion symmetrization makes the control bidirectional but erases the forward orientation; product-ratio mixing creates one monoid-compositional channel together with mixed-sign beat channels; and the noncommuting logarithmic drift turns the resulting coordinate alphabet into a dynamical control system.

group basis gauge $\rightarrow$ monoid atoms (primes) $\rightarrow$ symmetric instruction $\rightarrow$ code/beat mixing $\rightarrow$ noncommuting drift $\rightarrow$ sector control.

This interpretation should not be confused with a physical discovery. Mathematically, prime selection is obtained only after M_P is included as distinguished structure. Physically, the central question is therefore sharper: does any natural system supply both a multiplicative state space and a preferred forward/preparation monoid whose atoms are prime-indexed?

12. Falsifiability and the Prime-Selection Problem

The identity $Q_x Q_y = \frac{1}{2}(Q_{\{xy\}} + Q_{\{x/y\}})$ can be written for arbitrary commuting multiplicative shifts, not only primes. The group alone also has a $GL(m, \mathbb{Z})$ basis gauge. Thus neither multiplicative mixing nor group controllability, by itself, establishes physical prime privilege. In the revised model the mathematical selector is the distinguished monoid M_P : its atoms are prime axes. A physical prime-selection claim must therefore show that an analogous forward/preparation structure is actually present, rather than inferred after the fact.

- Oriented primitive addressability: experimentally independent forward operations must define a discrete nonnegative-exponent monoid, not merely an unoriented group basis.
- Atomic factorization: forward composite responses must factor into the monoid atoms without additional primitive generators.
- Code-beat asymmetry: forward-only composition $S_p S_q$ produces the product direction pq , whereas the ratio direction p/q requires an inverse-direction path and appears under symmetric/inversion-enabled mixing.
- Null comparison: arbitrary irrational or continuously tunable scale directions must not furnish an equally natural oriented atomic monoid with the same operational status.

A particularly direct operational test is to compare oriented and symmetric controls. If inverse-direction operations are selectively disabled, the algebra predicts retention of the product path $S_{\{pq\}} = S_p S_q$ but removal of the ratio path $S_{\{p/q\}} = S_p S_q^{-1}$. Failure to observe any meaningful distinction between forward composition and mixed-sign beat channels would count against the proposed physical interpretation, while leaving the mathematics intact.

13. Relation to Existing Mathematical Lineages

The prime-exponent coordinate map is not introduced here as a new representation. It belongs to the same broad arithmetic-coordinate lineage as the Bohr lift, where prime factorization converts Dirichlet monomials into multi-indices on infinitely many variables [1,2]. The lattice dynamics likewise sits naturally beside continuous-time quantum walks on graphs and integer lattices [3,4]. Finally, the use of dynamical Lie algebras and connectivity to establish finite-dimensional quantum controllability is standard in quantum control theory [5–7].

The contribution of the present preprint is the particular synthesis studied here: a multiplicative prime-exponent state space with an explicit distinction between group-basis gauge and monoid-level prime selection; inversion-symmetric prime controls with orientation loss; product-ratio and hypercube mixing together with the code-beat split; a quadratic logarithmic drift with a prime-pair bare splitting; and a general parity-sector controllability theorem. No historical priority claim is made for the individual ingredients.

14. Limitations and Open Problems

- Physical realization is open. No natural system is identified here as a prime-control lattice.
- The exact product-ratio algebra is an infinite-lattice statement; hard-wall truncation introduces boundary corrections to operator products.
- The sector controllability theorem assumes independently addressable prime-axis controls. Constrained or collective controls require separate analysis.
- Robustness to noise, decoherence, control bandwidth, and approximate rather than exact logarithmic coordinates is not analyzed.
- Physical prime selection remains an empirical barrier. The group alone does not select primes; a natural realization must supply a distinguished forward/preparation monoid analogous to M_P .
- Symmetric controls satisfy $Q_x = Q_{\{1/x\}}$ and therefore erase orientation. Recovering a physically meaningful forward direction requires extra structure beyond the symmetric dynamics.
- Infinite-dimensional controllability is not claimed; the theorem is finite-dimensional for every fixed K and finite prime set.

15. Conclusion

Prime-Control Lattices provide a minimal mathematical universe in which unique factorization becomes a coordinate architecture only after two levels are separated. The multiplicative group supplies a free-abelian state lattice but admits a $GL(m, \mathbb{Z})$ basis gauge; the distinguished nonnegative-exponent monoid removes that ambiguity by selecting primes as its atoms, uniquely up to permutation. Symmetric prime shifts then obey exact product-ratio and hypercube mixing identities, with one forward compositional code channel and mixed-sign beat channels. A quadratic logarithmic drift makes the symmetric controls noncommuting and gives an exact bare product-versus-ratio splitting $\Omega \log p \log q$. Finite hard-wall truncations remain fully unitary controllable inside each inversion-parity sector for any finite set of distinct primes.

The model therefore demonstrates a more precise statement than the original group-only formulation: a multiplicatively encoded universe can support a complete prime-indexed control architecture if it carries not only the multiplicative group but also a distinguished factorization orientation whose atoms are prime directions. Inversion-symmetric dynamics then preserves control power while losing that orientation, making the positive/preparation structure an independently testable ingredient. Whether the physical universe contains a subsystem with these properties remains an empirical question.

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Appendix A. Edge-Generator Lemma Used in Theorem 4

For completeness, let $D = \text{diag}(E_1, \dots, E_d)$ and let $X_{ab} = |a\rangle\langle b| + |b\rangle\langle a|$ on an edge $a-b$. Then $\text{ad}_D^2(X_{ab}) = (E_a - E_b)^2 X_{ab}$ up to the conventional sign induced by skew-Hermitian generators. If a control is a sum of edge terms whose squared gaps are distinct, a polynomial in ad_D^2 can interpolate one selected edge term and annihilate the others. Commuting the isolated symmetric term with D produces the antisymmetric edge term. For a connected graph, such pairs on all edges generate $\mathfrak{su}(d)$: edge commutators give diagonal differences, and commutators along paths generate non-neighbor couplings. This is the elementary graph-Lie mechanism used in the proof.

Appendix B. Suggested Citation

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